

Principle of Inheritance and Variation

1. The ratio of total number of genotypes and phenotypes for ABO blood types in human being is

- (1) 3 : 2
- (2) 4 : 1
- (3) 3 : 1
- (4) 2 : 1

2. Which of the following is not a correct dominant recessive trait pairs of *Pisum sativum*?

- (1) Axial-terminal flower position, Tall-dwarf stem height.
- (2) Yellow-green pod colour, round-wrinkled seed shape
- (3) Full-constricted pod shape, Yellow-green seed colour
- (4) Violet-white flower colour, Inflated-constricted pod shape

3. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R).

Assertion (A) : In human, we don't just have tall or short people as two distinct alternatives but a whole range of possible heights.

Reason (R) : Human skin colour is a classic example of pleiotropic gene.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
- (2) (A) is correct but (R) is not correct.
- (3) (A) is not correct but (R) is correct.
- (4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

4. Polygenic traits

- (1) Are generally controlled by single gene with more than 2 alleles
- (2) Such as human skin colour are generally controlled by 3 or more genes
- (3) Have only two distinct alternate forms
- (4) Are seen in humans only

5. Find the incorrect statement w.r.t. Chromosomal mapping :

- (1) Crossing over is important in locating genes on chromosome
- (2) Recombination frequency depends upon the distance between the genes
- (3) Recombination frequency is inversely proportional to distance between genes
- (4) The sequences and the relative distances between various genes is graphically represented in terms of recombination frequencies

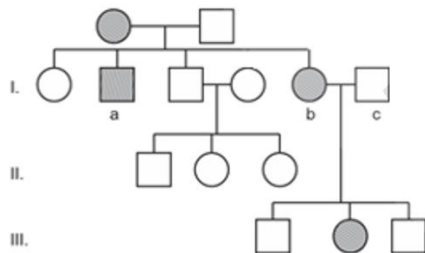
6. Haplodiploid method of sex determination is found in

- (1) Butterfly
- (2) Honey bee
- (3) Grasshopper
- (4) Fruit fly

7. Choose the incorrect one for flower colour inheritance in *Antirrhinum*.

- (1) Plants with RR genotype produces red flowers
- (2) It shows incomplete dominance
- (3) When F1 plants (Rr) flowers are self pollinated, the genotype and phenotype ratio are same in F2 generation
- (4) The genotype rr produces pink coloured flowers

8. A pedigree chart is given below. Identify incorrect option regarding this pedigree.



- (1) This pedigree can be applicable for myotonic dystrophy
 (2) Above pedigree can be exemplified by a genetic disorder like thalassemia
 (3) Male (I c) can be carrier if the trait is considered for colour-blindness
 (4) This pedigree is not applicable for Y linked disorders
9. Which of the given crosses produces tall and dwarf pea plant in equal proportion?

- (1) $TT \times tt$
 (2) $Tt \times tt$
 (3) $TT \times Tt$
 (4) $tt \times tt$
10. Read the following assertion and reason
Assertion : Law of segregation is based on the fact that alleles do not show any blending.
Reason : Law of dominance is used to explain the expression of only one parental character in a monohybrid cross in the F₁.
- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
 (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
 (3) Assertion is true statement but Reason is false
 (4) Both Assertion and Reason are false statements

11. Sex of the chicks is determined by
- (1) Their female parent
 (2) Both of their male and female parents
 (3) The temperature at which the egg is incubated
 (4) Their male parent

12. Identify the similarities between phenylketonuria and Down's syndrome amongst the following.
- a. Affected individuals show retarded mental development
 b. The disorders are associated with autosomes.
 c. Number of autosomes remains the same in affected and unaffected individuals.
 d. These disorders are due to failure of cytokinesis after telophase stage of cell division.

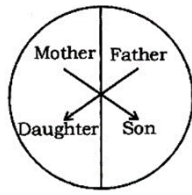
The correct ones are

- (1) (b) and (d) only
 (2) (b), (c) and (d) only
 (3) (a), (b) and (c) only
 (4) (c) and (d) only
13. In human, during each pregnancy there is always ____ probability of either a male or female child.
 Fill the blank with suitable option.

- (1) 75%
 (2) 50%
 (3) 25%
 (4) 100%
14. Select statement which is true for Punnett square.
- (a) Developed by a British geneticist.
 (b) Used to calculate the probability of all possible genotypes of offsprings.
 (c) Useful for monohybrid crosses only.
 (d) It is a mathematical formula which helps to calculate the types of gametes.

- (1) (a), (b) and (c)
 (2) (a) and (d)
 (3) (b) and (d)
 (4) Only (a) and (b)
15. The experimental verification of the chromosomal theory of inheritance was done by
- (1) Sutton and Boveri
 (2) Bateson
 (3) T.H Morgan and his colleagues
 (4) Hugo de Vries

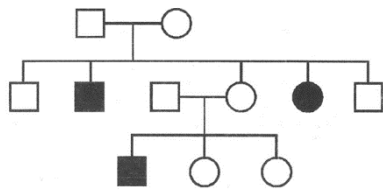
16. Represented below in the inheritance pattern of certain type of traits in humans. Which one of the following conditions could be an example of this pattern?



- (1) Haemophilia
(2) colour blindness
(3) Phenylketonuria
(4) Both (1) and (2)
17. Given below are two statements:
Statement-I : Modified allele could be responsible for production of normal enzyme.
Statement-II : Normal/unmodified allele is responsible for production of normal enzyme.
Choose the most appropriate answer from the options given below:
- (1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.
18. Select the incorrect statement for sickle cell anaemia.
- (1) It is autosomal recessive disorder
(2) It is caused by transversion mutation
(3) It is quantitative problem of synthesising too few globin molecules
(4) $Hb^S Hb^S$ individuals die before attaining maturity
19. Test cross is a cross between
- (1) Hybrid $\times$ Dominant parent ($Rr \times RR$)
(2) Hybrid $\times$ Recessive parent ($Rr \times rr$)
(3) Hybrid $\times$ Hybrid ($Rr \times Rr$)
(4) All the above

20. Identify the set of correct statement/s.
- (a) Nowadays genetic maps are extensively used as a starting point in genome sequencing in HGP.
(b) In Morgan's experiment's, wild type allele and mutant type allele are represented with (+) and (–) sign in superscript, respectively.
(c) Strength of linkage between Y and W is higher than W and M.
(d) Pleiotropy in most cases is the effect of a gene on metabolism pathways.
- (1) (a), (b) and (d) only
(2) (b), (c), (d) only
(3) (a) and (d) only
(4) (a), (b), (c) and (d) only
21. A normal man whose father was colour-blind marries a woman who had a colour-blind mother and normal father. What would be the probability of their male children to be colour-blind?
- (1) 0%
(2) 25%
(3) 50%
(4) 100%
22. In any cross, if recombination frequency is 5%, it means
- (1) The distance between the observed genes is 5 cM
(2) Number of recombinants is 5 in 10 progenies
(3) Their crossover value is $1/5$
(4) Out of total progeny produced, 5 are parental types
23. A pink snapdragon flower is crossed to a white snapdragon flower. What is the probability of getting a red snapdragon?
- (1) 1
(2) $1/2$
(3) $1/4$
(4) Zero

24. Study of pedigree chart. What does it show?



- (1) Inheritance of a condition like cystic fibrosis as an autosomal dominant trait
- (2) Inheritance of a recessive sex-linked disease like haemophilia
- (3) Inheritance of sex-linked inborn error of metabolism like phenylketonuria
- (4) Inheritance of a condition like sickle cell anaemia as an autosomal recessive trait

25. If gene B controls starch synthesis in garden pea, then its seeds with genotype Bb will be

- (1) Round with large starch grains
- (2) Round with intermediate size starch grains
- (3) Wrinkled with small starch grains
- (4) Wrinkled with large starch grains

26. Given below are two statements:

Statement-I : One gene affect many characters called pleiotropic gene.

Statement-II : Dominance is an autonomous feature of a gene or the gene product.

Choose the most appropriate answer

- (1) Both Statement-I and Statement-II are incorrect.
- (2) Statement-I is correct but statement-II is incorrect.
- (3) Statement-I is incorrect but Statement-II is correct.
- (4) Both Statement-I and Statement-II are correct.

27. In which of the following disorders, affected individual possesses 45 chromosomes?

- (1) Turner's syndrome
- (2) Down's syndrome
- (3) Klinefelter's syndrome
- (4) Hemophilia

28. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R).

Assertion (A) : Maximum 50% non-parental gene combinations are possibly produces during gamete formation.

Reason (R) : Maximum frequency of crossing over could be 100% because all four chromatids of tetrad are involves in crossing over.

In the light of the above statements, choose the correct answer from the options given below:

- (1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
- (2) (A) is correct but (R) is not correct.
- (3) (A) is not correct but (R) is correct.
- (4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

29. The individuals with trisomy of 21st chromosome are characterised by all following features, except

- (1) Round head
- (2) Broad palm with characteristic palm crease
- (3) Furrowed tongue
- (4) Rudimentary ovaries

30. Colour blindness is a/an

- (1) X-linked recessive disorder
- (2) X-linked dominant disorder
- (3) Autosomal recessive disorder
- (4) Autosomal dominant disorder

31. Select the correct set of parents those can produce child with blood group 'O'.

- a. $I^A i \times I^A i$
- b. $I^B i \times I^A I^A$
- c. $ii \times I^A i$
- d. $I^B I^B \times I^B i$

- (1) a and c
- (2) b and d
- (3) a and d
- (4) b and c

32. The term used by Morgan to describe the physical association of genes on a chromosome is

- (1) Linkage
- (2) Dominance
- (3) Inheritance
- (4) Recombination

33. Read the statements & Select the correct ones w.r.t. Morgan's experiments on linkage

- (A) When genes are grouped on the same chromosome, some genes were loosely linked and showed very low recombination.**
- (B) The genes for white eye colour and yellow body colour were tightly linked and showed 1.3 percent recombination.**
- (C) The genes for white eye colour and miniature wings were loosely linked and showed 37.2% parental types.**
- (D) Morgan found that proportion of parental gene combination was much higher than non-parental gene combination.**

- (1) Statements (A) & (C)
- (2) Statements (B), (C) & (D)
- (3) Statement (B) only
- (4) Statements (B) & (D)

34. In pea plants, yellow seeds are dominant to green. If a heterozygous yellow seeded plant is crossed with a green seeded plant, what ratio of yellow and green seeded plants would you expect in F1 generation?

- (1) 3 : 1
- (2) 15 : 15
- (3) 9 : 1
- (4) 1 : 3

35. Punnett square was developed by

- (1) British Zoologist, Reginald C. Punnett
- (2) German Botanist, Reginald C. Punnett
- (3) Stanford Geneticist Reginald C. Punnett
- (4) British Geneticist Reginald c. Punnett

36. Identify the set of correct statement :

- (a) Mendel investigated inheritance with usage of statistical analysis and mathematical logics.**
- (b) Mendel's experiments had a small sampling size.**
- (c) Mendel conducted artificial pollination for the hybridisation experiment.**
- (d) Mendel selected 14 true-breeding pea plant varieties.**

- (1) (a) and (c) only
- (2) (b), (c) and (d) only
- (3) (a), (c) and (d) only
- (4) (a), (b), (c) and (d) all

37. Select incorrect one (w.r.t. reciprocal cross)

- (1) To know whether the alleles are present on sex chromosomes or autosomes
- (2) It is mad to eliminate the effect of nuclear traits
- (3) two individuals with contrast genotypes are involved
- (4) Results are not changed for autosomal traits

38. Which of the following phenotypic ratio was found by Mendel in F2 generation of dihybrid cross :

- (1) 1 : 2 : 1
- (2) 1 : 2 : 1 : 2 : 4 : 2 : 1 : 2 : 1
- (3) 9 : 3 : 3 : 1
- (4) 1 : 1 : 1 : 1

39. Mendel selected Pisum sativum for hybridisation experiments because of :

- (1) Clear contrasting character and short life span
- (2) Long life span and non - fertile hybrids
- (3) Presence of unisexual flowers
- (4) Infertile hybrids and production of large number of seeds by each plant

40. Mark the odd one (w.r.t. ture breeding line)

- (1) Shows the stable trait inheritance
- (2) Shows expression for few generations only
- (3) Undergone continuous self-pollination
- (4) both (1) & (3)

41. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R).

Assertion (A) : Phenylketonuria is an example of pleiotropy.

Reason (R) : Phenylketonuria manifests mental retardation and a reduction in hair and skin pigmentation besides too much phenylalanine in the body.

Choose the correct answer from the options

- (1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
- (2) (A) is correct but (R) is not correct.
- (3) (A) is not correct but (R) is correct.
- (4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

42. A marriage between normal visioned man and colour blind woman will produce offspring.

- (1) colour blind sons and 50% carrier daughters
- (2) 50% colour-blind sons and 50% carrier daughters
- (3) Normal males and carrier daughters
- (4) colour blind sons and carrier daughters

43. Given below are two statements:

Statement-I : Heterozygous female for haemophilia may transmit the disease to sons.

Statement-II : Sickle-cell anaemia is caused by the substitution of valine (val) by glutamic acid (Glu) of β -globin chain.

Choose the correct answer from the options

- (1) Both Statement-I and Statement-II are incorrect.
- (2) Statement-I is correct but statement-II is incorrect.
- (3) Statement-I is incorrect but Statement-II is correct.
- (4) Both Statement-I and Statement-II are correct.

44. Identify the set of incorrect statement/s.

- (a) Colour blindness is a sex-linked dominant disorder.
- (b) Colour blindness occurs in about 0.4 per cent of males and 8 per cent of females.
- (c) Male are in heterozygous condition for haemophilia and colour blindness.
- (d) Colour blindness is due to absence of some specific proteins which forms cones.

Choose the correct answer

- (1) only (a)
- (2) (a), (b) and (c) only
- (3) (a) and (b) only
- (4) only (d)

45. The phenotype of an individual may be affected if the modified allele produces :

- (a) No enzyme at all
- (b) The normal/less efficient enzyme
- (c) A non-functional enzyme

- (1) Only (a) is correct
- (2) (a) and (c) are correct
- (3) (b) and (c) are correct
- (4) Only (c) is correct

46. Mark the odd one (w.r.t. F₂ generation of Mendelian dihybrid cross)

- (1) Frequency of TtRR genotype = 25.0%
- (2) Frequency of ttRR genotype = 6.25%
- (3) Frequency of TTRR genotype = 6.25%
- (4) Frequency of ttRr genotype = 12.5%

47. If a pea plant produces 1280 seeds during a dihybrid cross between round-yellow and wrinkled-green plant. Then how many seed are wrinkled-yellow, round-yellow and wrinkled green respectively

- (1) 640, 480, 1280
- (2) 240, 720, 80
- (3) 640, 1280, 320
- (4) 160, 1440, 480

48. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R).

Assertion (A) : Law of segregation is based on the fact that the alleles do not show any blending.

Reason (R) : Factor or alleles of a pair segregate from each other such that a gamete receives only one of the two factor.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
- (2) (A) is correct but (R) is not correct.
- (3) (A) is not correct but (R) is correct.
- (4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

49. Identify the set of incorrect statement/s.

(a) Sometime, F₁ do not resemble either of two parents for flower colour in some plants.

(b) Occasionally a single gene product may produce more than one phenotype.

(c) After maturation of seeds, BB seeds are round and Bb seeds are wrinkled.

(d) Dominance is an autonomous feature of a gene.

Choose the correct answer from the option given below :

- (1) (c) only
- (2) (a), (c) and (d) only
- (3) (c) and (d) only
- (4) (b) and (c) only

50. Given below are two statements:

Statement-I : Mendel's study found that the factors independently assort and combine in all permutations and combinations.

Statement-II : Later, it was found that mendel's law of independent assortment does not hold true for the genes located on the same chromosomes.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Statement-I and Statement-II are incorrect.
- (2) Statement-I is correct but statement-II is incorrect.
- (3) Statement-I is incorrect but Statement-II is correct.
- (4) Both Statement-I and Statement-II are correct.

Answer Key

1. (1)
2. (2)
3. (2)
4. (2)
5. (3)
6. (2)
7. (4)
8. (3)
9. (2)
10. (2)
11. (1)
12. (3)
13. (2)
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15. (3)
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17. (4)
18. (3)
19. (2)
20. (1)
21. (3)
22. (1)
23. (4)
24. (4)
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26. (2)
27. (1)
28. (2)
29. (4)
30. (1)
31. (1)
32. (1)
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34. (2)
35. (4)
36. (3)
37. (2)
38. (3)
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40. (2)
41. (4)
42. (4)



- 43. (2)
- 44. (2)
- 45. (2)
- 46. (1)
- 47. (2)
- 48. (4)
- 49. (3)
- 50. (4)

